**Joule, Volume *8***

## Supplemental information

# Scalability and stability in $CO_2$ reduction via tomography-guided system design

**Colin P. O'Brien, David McLaughlin, Thomas Böhm, Yurou Celine Xiao, Jonathan P. Edwards, Christine M. Gabardo, Markus Bierling, Joshua Wicks, Armin Sedighian Rasouli, Jehad Abed, Daniel Young, Cao-Thang Dinh, Edward H. Sargent, Simon Thiele, and David Sinton**

Table S1 Specifications and comparison of the commercial gas diffusion layers.

| Gas diffusion layer | GDL240 | H23C8 | GDS2230 | 25BC |
|---|---|---|---|---|
| Manufacturer | CeTech | Freudenberg | AvCarb | Sigracet |
| Thickness (µm) | 240 | 230 | 275 | 235 |
| Substrate material | Carbon fiber paper | Carbon fiber paper | Carbon fiber paper | Carbon fiber paper |
| Hydrophobic MPL | Yes | Yes | Yes | Yes |
| Area weight (g/m$^2$) | 90 | 135 | 99 | 86 |
| Through plane electrical area-specific resistance (mΩ cm$^2$) | <15 | 8 | 10 | <12 |

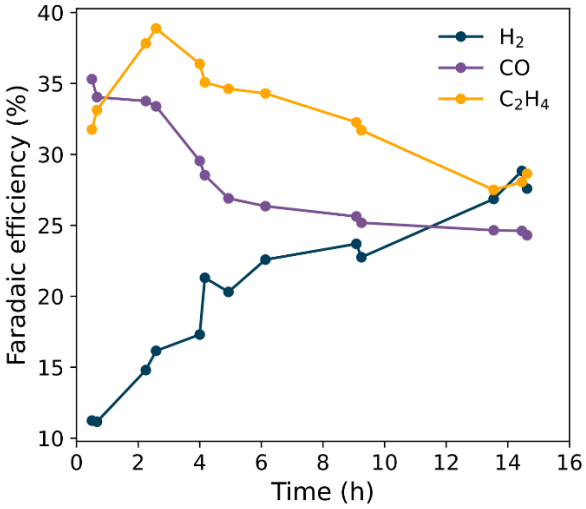


Figure S1 Stability of the gas product selectivity of the H23C8 GDL at 200 mA/cm$^2$ without periodic washing.

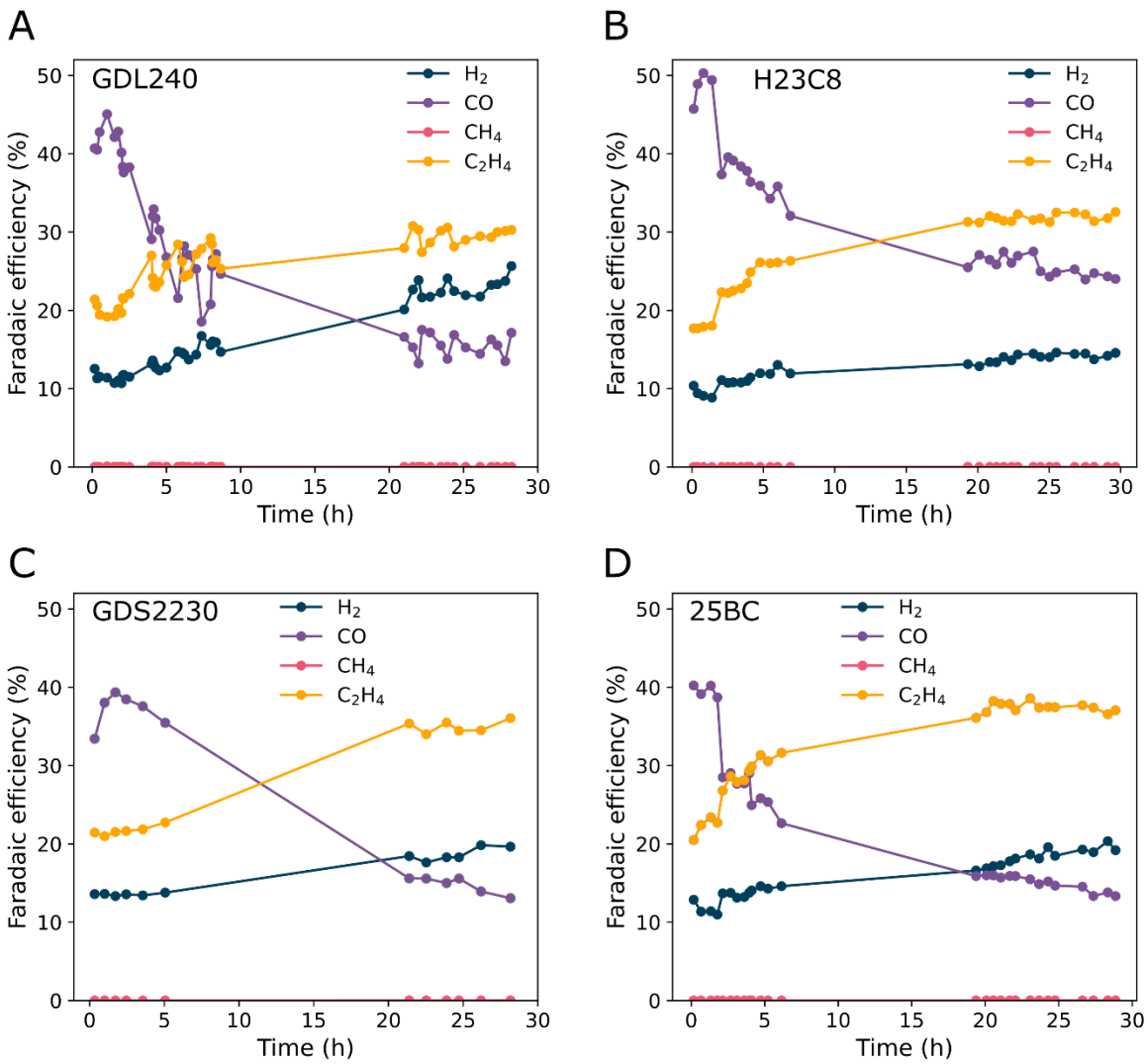


Figure S2 Stability of the commercial GDLs at 200 mA/cm$^2$ with the FE to each gas product. The stability is shown for (A) GDL240, (B) H23C8, (C) GDS2230, and (D) 25BC.

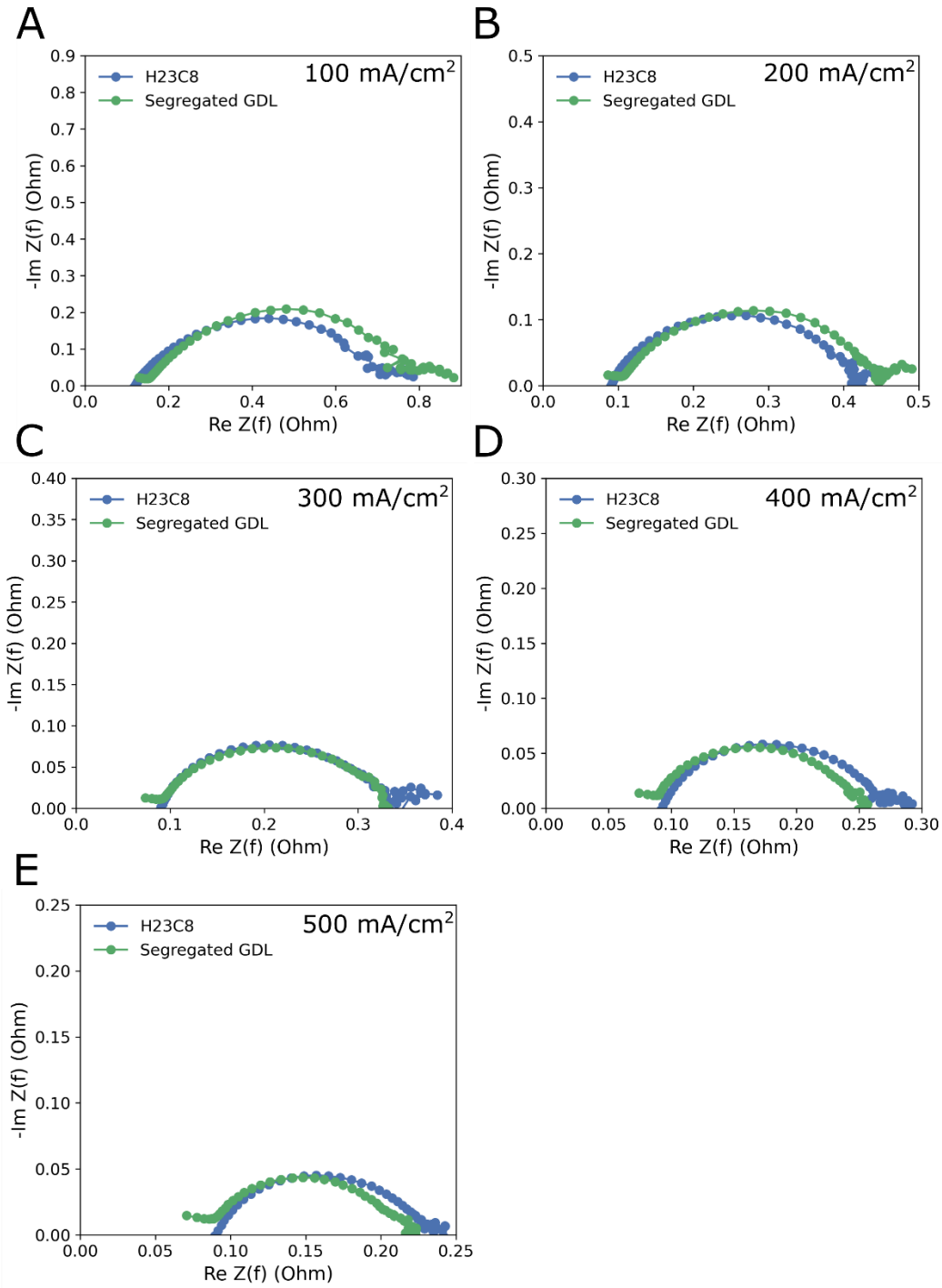


Figure S3 Nyquist plots of the segregated GDL compared to the Freudenberg H23C8 commercial GDL. The Nyquist plots are shown for (A) 100 mA/cm$^2$, (B) 200 mA/cm$^2$, (C) 300 mA/cm$^2$, (D) 400 mA/cm$^2$, and (E) 500 mA/cm$^2$. The electrochemical impedance spectroscopy was conducted with frequencies from 100,000 to 0.1 Hz with an amplitude of 4% of the applied current.

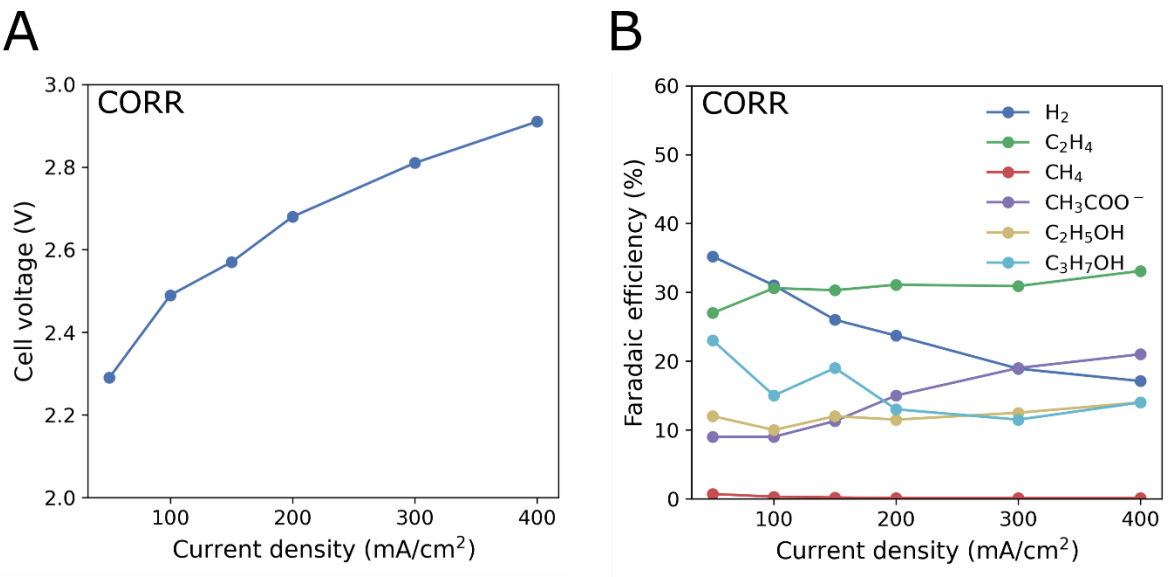


Figure S4 Performance of the segregated GDL for the CO reduction reaction (CORR). (A) the cell voltage and (B) the product selectivity of the segregated GDL while performing the CORR. 1M KOH was used as the anolyte.

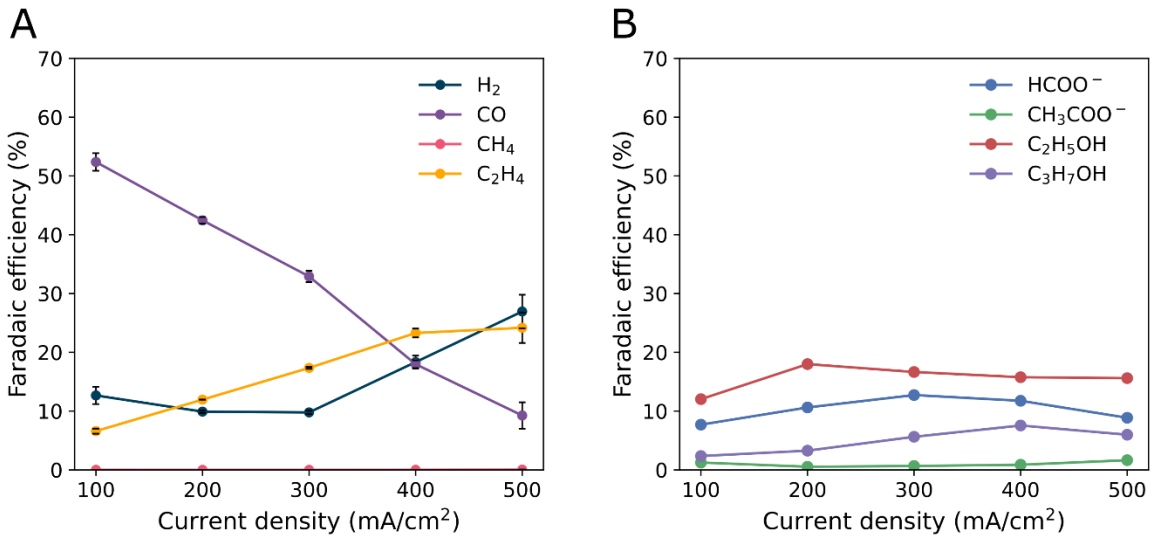


Figure S5 Full product selectivity of the segregated GDL. The (A) gas products and (B) liquid products at each current density. Error bars represent the standard deviation of at least three measurements. The liquid products were collected over more than 1 h at each current density.

Table S2 Transport parameters of the segregated GDE for all directions

| Direction | Laplace Diffusion | Knudsen Diffusion | Conductivity | Permeability |
|---|---|---|---|---|
| | % | % | % | $m^{-2}$ |
| **Catalyst Layer** | | | | |
| *Normal Direction (z)* | 51.19 | 37.05 | 7.19 | 7.72E-16 |
| In-plane Direction (X) | 49.29 | 34.97 | 6.47 | 6.87E-16 |
| In-plane Direction (y) | 58.61 | 48.54 | 10.99 | 9.92E-16 |
| **MPL** | | | | |
| *Normal Direction (z)* | 19.96 | 16.78 | 1.77 | 1.71e-15 |
| In-plane Direction (X) | 32.40 | 27.23 | 2.88 | 2.79e-15 |
| In-plane Direction (y) | 28.84 | 24.23 | 2.56 | 2.49e-15 |
| **Porous Carbon** | | | | |
| *Normal Direction (z)* | 61.92 | 54.71 | 5.29 | 6.00e-15 |
| In-plane Direction (X) | 58.09 | 45.44 | 4.42 | 4.39e-15 |
| In-plane Direction (y) | 64.59 | 54.98 | 6.70 | 5.71e-15 |
| **Carbon Paper** | | | | |
| *Normal Direction (z)* | 48.74 | n/A | 2.78 | 5.40e-15 |
| In-plane Direction (X) | 47.45 | n/A | 2.23 | 6.61e-15 |
| In-plane Direction (y) | 50.66 | n/A | 2.89 | 9.01e-15 |

Table S3 Transport parameters with errors for those volumes that were calculated from several sub volumes. The errors represent statistical variations and do not take into account uncertainties in the segmentation or computation processes.

| Direction | Laplace Diffusion | Knudsen Diffusion | Conductivity | Permeability |
|---|---|---|---|---|
| | % | % | % | $m^{-2}$ |
| **Catalyst Layer** | | | | |
| *Normal Direction (z)* | 51.19 ± 3.34 | 37.05 ± 3.95 | 7.19 ± 1.12 | 7.72 ± 2.04 e-16 |
| In-plane Direction (X) | 49.29 ± 3.29 | 34.97 ± 3.71 | 6.47 ± 1.15 | 6.87 ± 1.76 e-16 |
| In-plane Direction (y) | 58.61 ± 2.77 | 48.54 ± 3.10 | 10.99 ± 2.01 | 9.92 ± 3.24 e-16 |
| **MPL** | | | | |
| *Normal Direction (z)* | 19.96 ± 0.93 | 16.78 ± 0.78 | 1.77 ± 0.08 | 1.71 ± 0.08 e-15 |
| In-plane Direction (X) | 32.40 ± 1.73 | 27.23 ± 1.45 | 2.88 ± 0.15 | 2.79 ± 0.15 e-15 |
| In-plane Direction (y) | 28.84 ± 1.34 | 24.23 ± 1.12 | 2.56 ± 0.12 | 2.49 ± 0.12 e-15 |
| **Porous Carbon** | | | | |
| *Normal Direction (z)* | 61.92 ± 3.56 | 54.71 ± 4.09 | 5.29 ± 1.82 | 6.00 ± 1.24 e-15 |
| In-plane Direction (X) | 58.09 ± 4.96 | 45.44 ± 5.96 | 4.42 ± 1.10 | 4.39 ± 0.92 e-15 |
| In-plane Direction (y) | 64.59 ± 2.86 | 54.98 ± 1.88 | 6.70 ± 1.75 | 5.71 ± 1.60 e-15 |

Table S4 Effective transport parameters of the segregated GDE for all directions. Self-diffusion of $CO_2$ is considered as the reference process. The Bosanquet diffusivities are given for Knudsen numbers of 0.35 for the CL and 0.14 for the porous carbon. Effective conductivities are calculated for reference conductivities of 2e5 S $m^{-1}$ for carbon and 5.81e7 S $m^{-1}$ for copper

| Direction | Laplace Diffusion | Knudsen Diffusion | Bosanquet Diffusion | Conductivity |
|---|---|---|---|---|
| | $m^2\ s^{-1}$ | $m^2\ s^{-1}$ | $m^2\ s^{-1}$ | S $m^{-1}$ |
| **Catalyst Layer** | | | | |
| *Normal Direction (z)* | 4.35e-06 | 2.63e-05 | 3.52e-06 | 6.63e+06 |
| In-plane Direction (X) | 4.19e-06 | 1.89e-05 | 2.82e-06 | 3.91e+06 |
| In-plane Direction (y) | 4.98e-06 | 2.63e-05 | 3.52e-06 | 6.63e+06 |
| **MPL** | | | | |
| *Normal Direction (z)* | 1.70e-06 | 9.09e-06 | 1.45e-06 | 3.55e+03 |
| In-plane Direction (X) | 2.75e-06 | 1.47e-05 | 2.35e-06 | 5.76e+03 |
| In-plane Direction (y) | 2.45e-06 | 1.31e-05 | 2.09e-06 | 5.13e+03 |
| **Porous Carbon** | | | | |
| *Normal Direction (z)* | 5.26e-06 | 2.96e-05 | 4.52e-06 | 1.06e+04 |
| In-plane Direction (X) | 4.94e-06 | 2.46e-05 | 4.17e-06 | 8.83e+03 |
| In-plane Direction (y) | 5.49e-06 | 2.98e-05 | 4.69e-06 | 1.34e+04 |
| **Carbon Paper** | | | | |
| *Normal Direction (z)* | 4.14e-06 | n/A | n/A | 5.56e+03 |
| In-plane Direction (X) | 4.03e-06 | n/A | n/A | 4.46e+03 |
| In-plane Direction (y) | 4.31e-06 | n/A | n/A | 5.78e+03 |

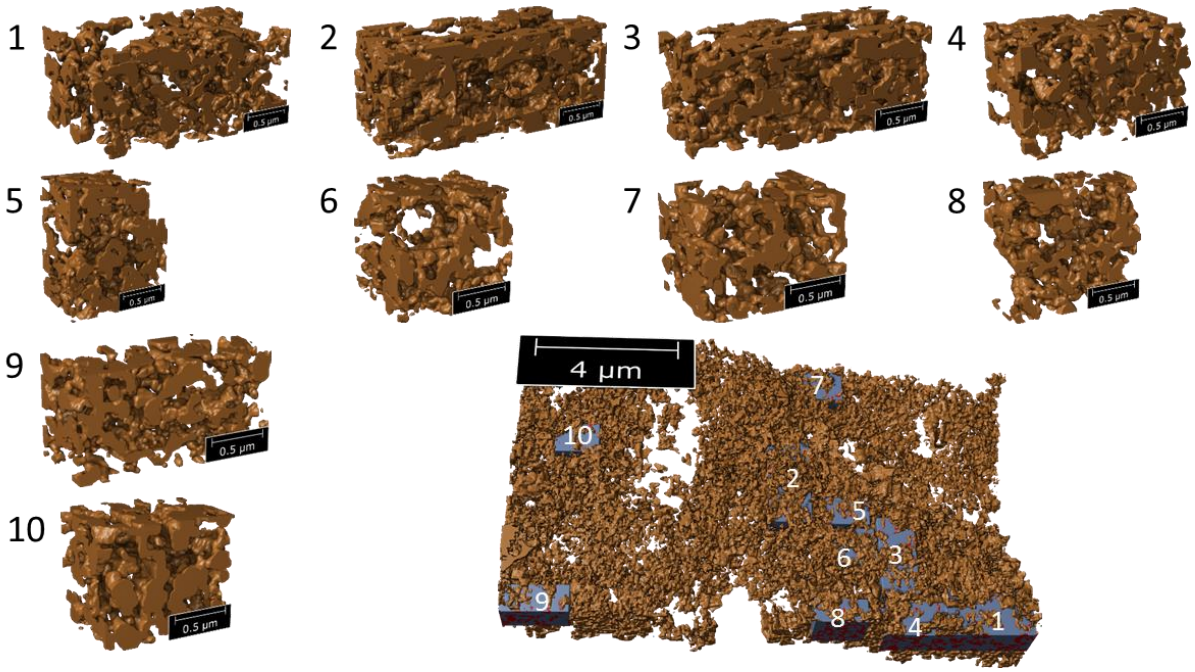


Figure S6 Volumes of the catalyst layer of the segregated GDE selected for analysis.

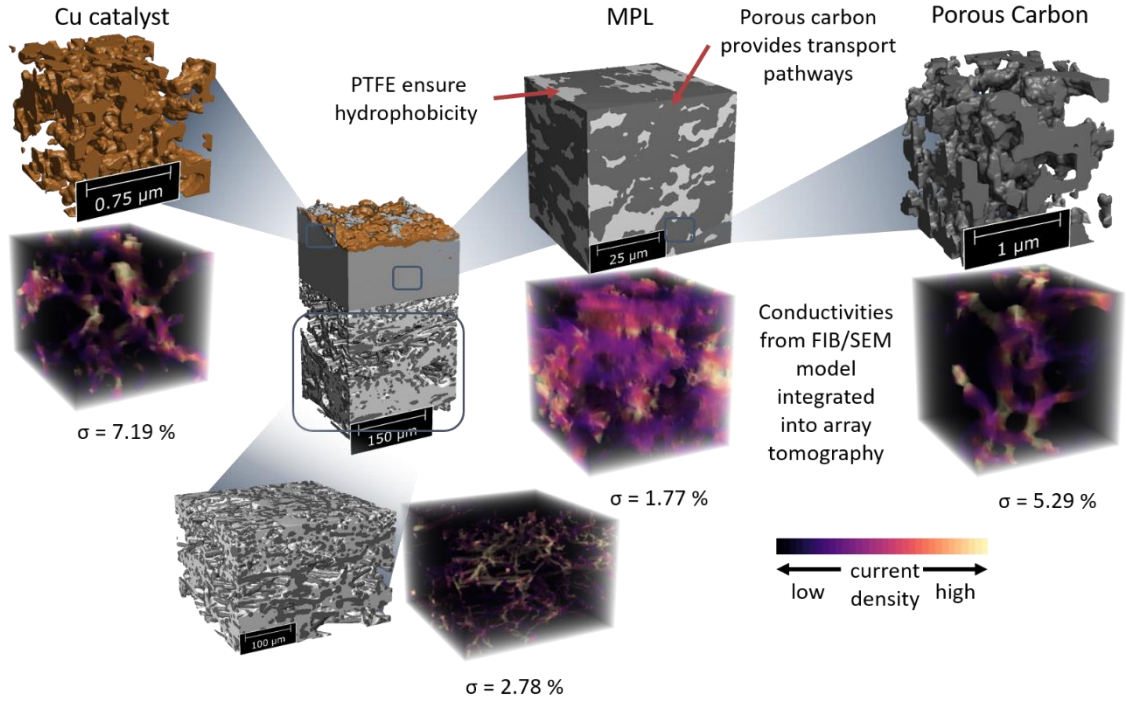


Figure S7 Illustration of the simulation of the conductivity parameters of the segregated GDE. σ, conductivity.

Table S5 Tomographies featured in this work. Primary volumes (in bold) and volumes used for calculations (in cursive) are listed. Visualizations are often cropped from the listed volumes to enable a good graphical representation. FIB/SEM tomography and array tomography are in general anisotropic for technical reasons. The two listed resolutions correspond to image resolution (in-plane resolution) and cutting distance (through-plane resolution).

| Volumes | Resolution | Dimension [Voxels] | Dimension [µm] |
|---|---|---|---|
| **FIB/SEM tomography** | 4/12.41 nm | 3601 x 1819 x 694 | 14.40 x 7.28 x 8.61 |
| *Cu* | 8 nm | 1800 x 585 x 1070 | 14.40 x 4.68 x 8.56 |
| *Cu transport (x10)* | 8 nm | various | various |
| *Porous carbon* | 8 nm | 614 x 600 x 500<br>750 x 396 x 428<br>550 x 396 x 500 | 4.91 x 4.80 x 4.00<br>6.0 x 3.17 x 3.42<br>4.4 x 3.17 x 4.0 |
| *FIB/SEM combined* | 8 nm | 1800 x 909 x 1070 | 14.4 x 7.27 x 8.56 |
| **EDX array tomography** | 400/400 nm | 1024 x 768 x 410 | 409.6 x 307.2 x 164 |
| *MPL volume (x4)* | 400 nm | 150 x 150 x 410 | 60 x 60 x 164 |
| **Micro CT** | 800 nm | 1860 x 1180 x 700 | 1488 x 944 x 560 |
| *Carbon paper* | 800 nm | 1860 x 1180 x 375 | 1488 x 944 x 300 |
| *MPL overview* | 800 nm | 1860 x 1180 x 325 | 1488 x 944 x 260 |

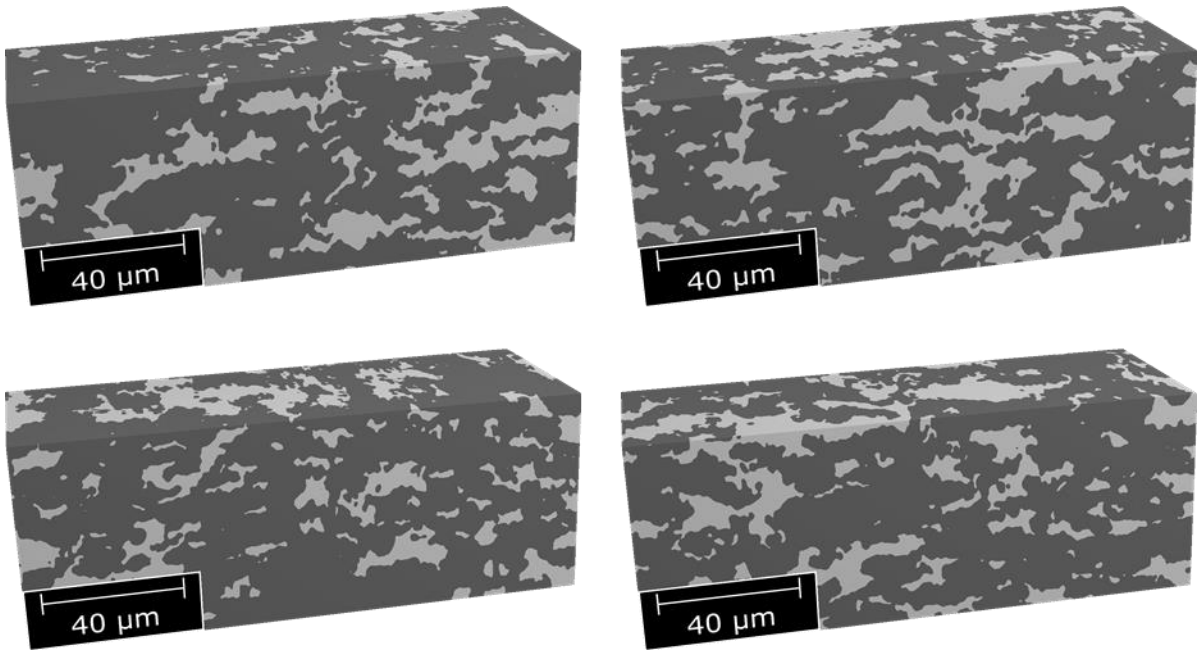


Figure S8 All four volumes of the MPL selected for analysis.

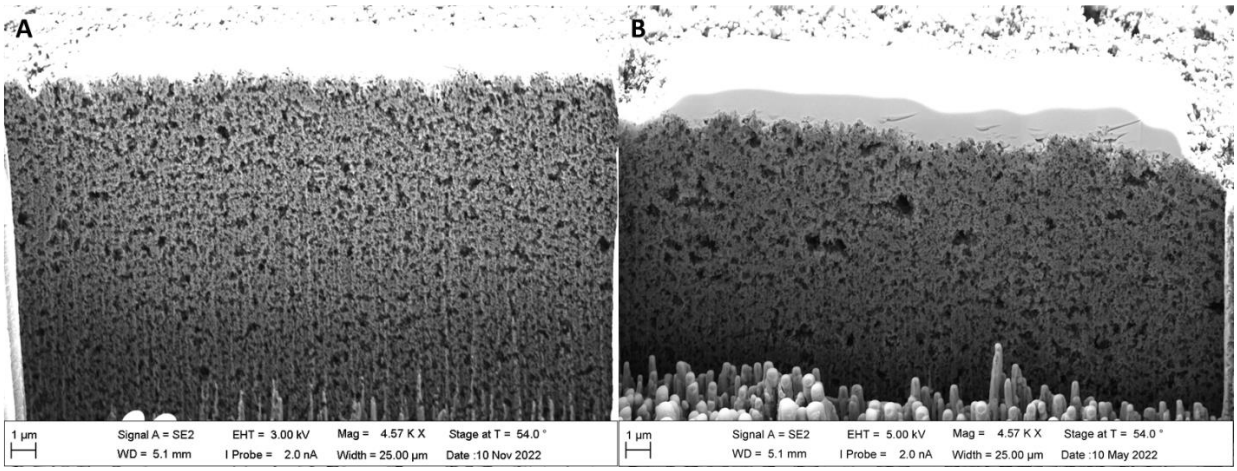


Figure S9 FIB cross sections of (A) Freudenberg H23C8 and (B) H23C2

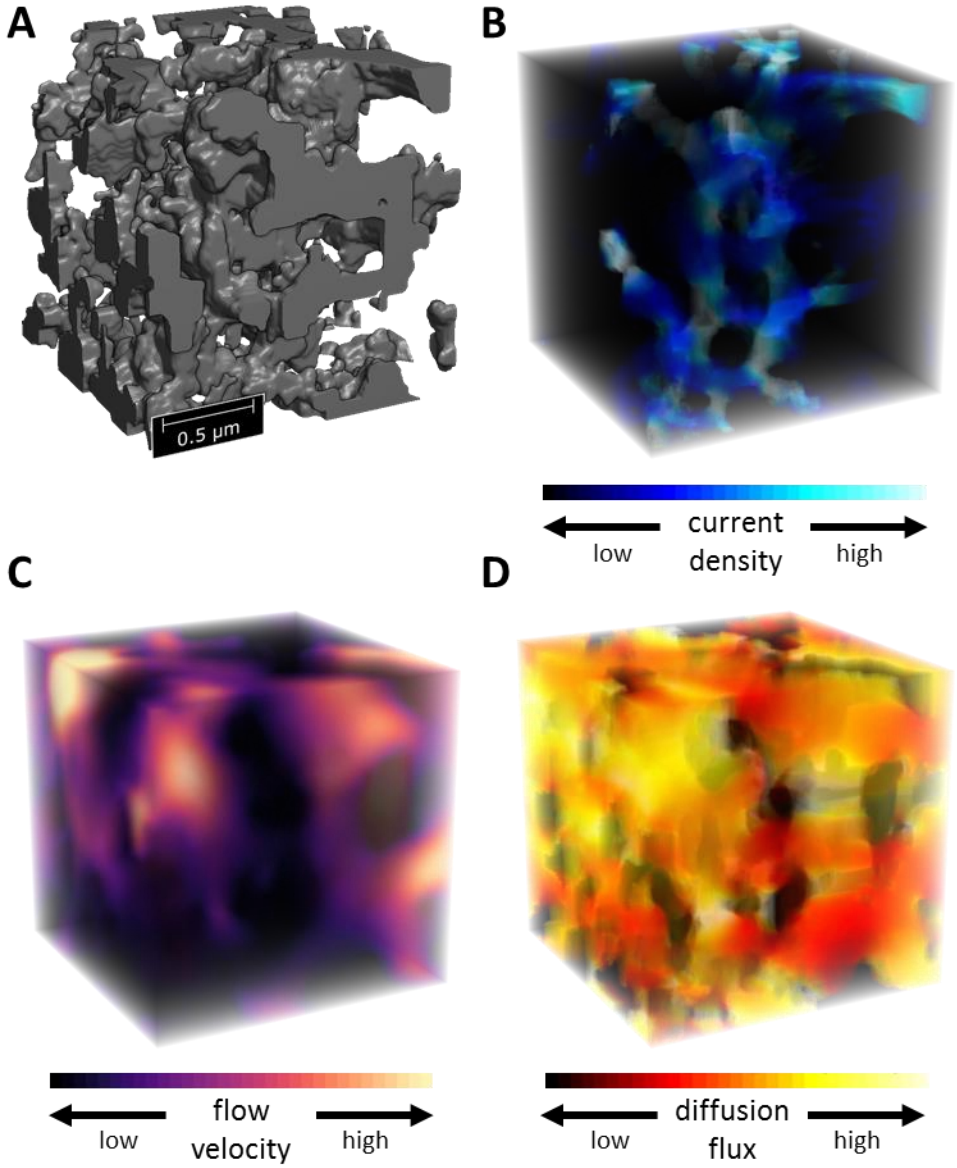


Figure S10 Qualitative visualization of transport processes in the porous carbon domain of the segregated MPL, with (A) structure, (B) conduction, (C) flow, and (D) diffusion.

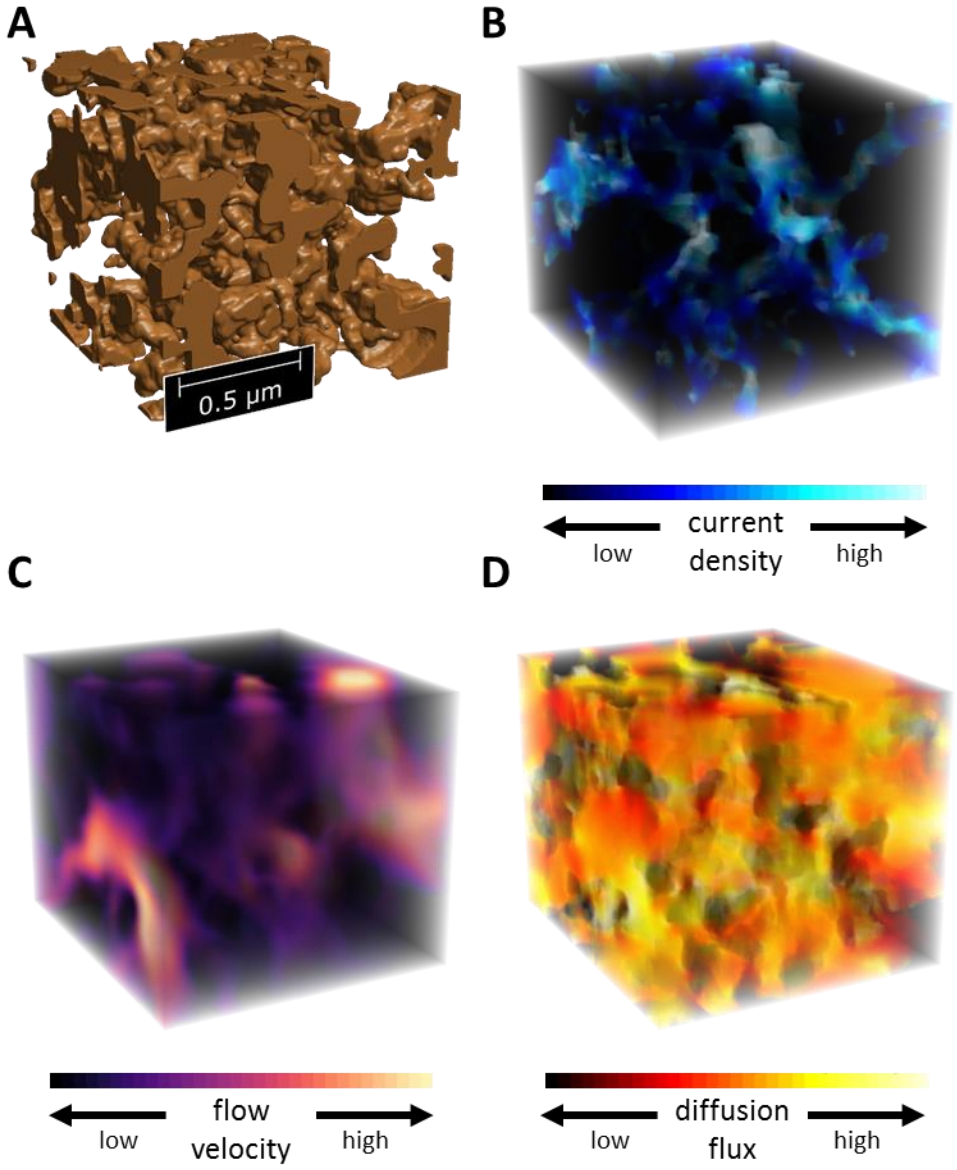


Figure S11 Qualitative visualization of transport processes in the copper catalyst layer of the segregated GDE, with (A) structure, (B) conduction, (C) flow, and (D) diffusion.

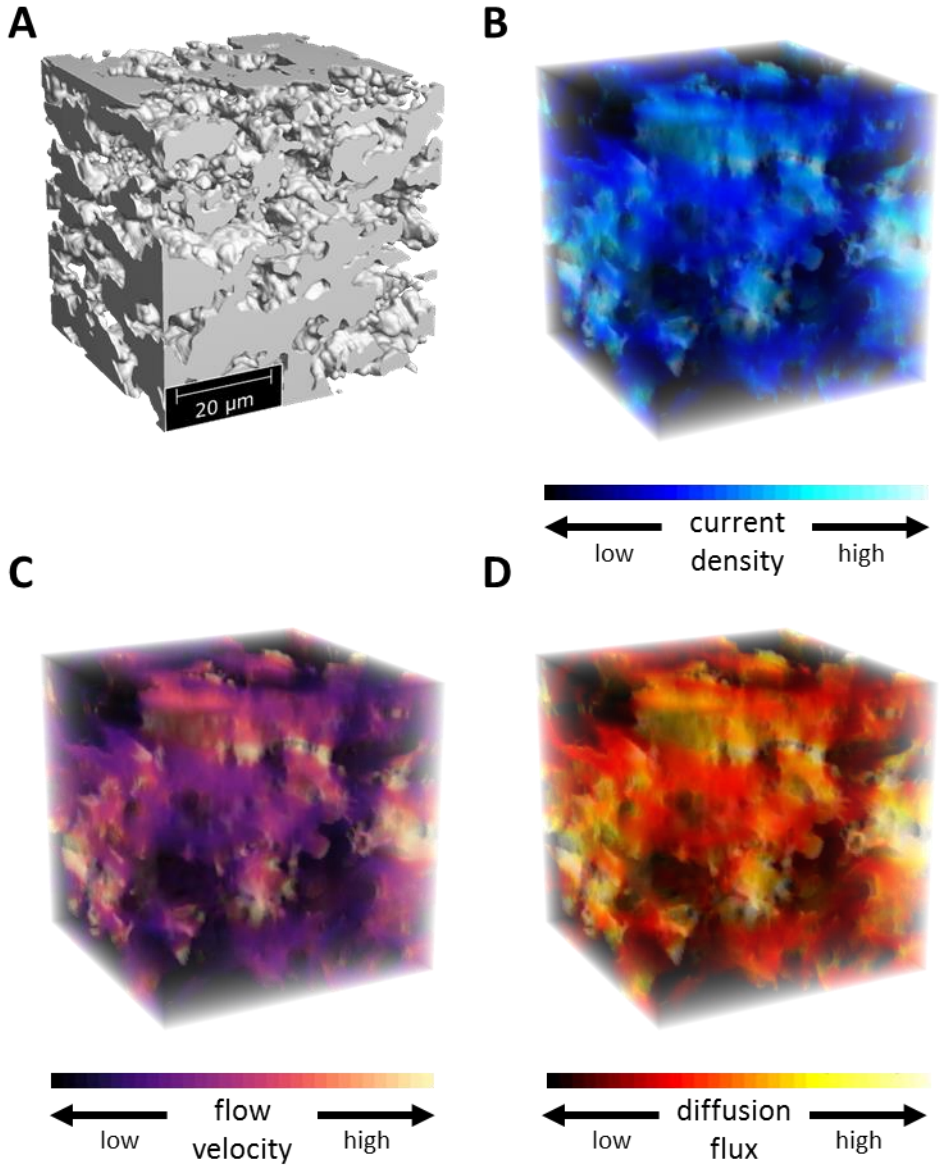


Figure S12 Qualitative visualization of transport processes in the two-domain MPL of the segregated GDE, with (A) PTFE structure, (B) conduction, (C) flow, and (D) diffusion.

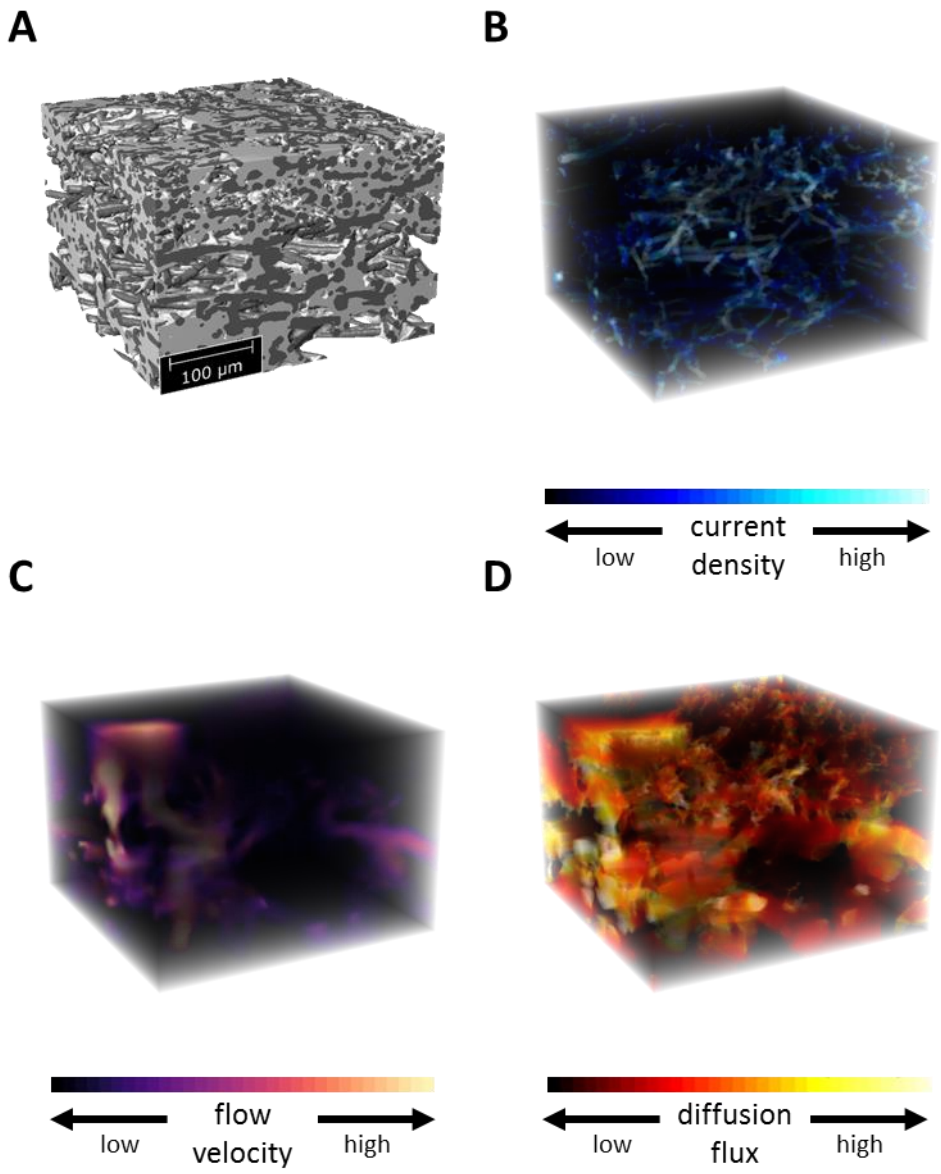


Figure S13 Qualitative visualization of transport processes in carbon paper of the segregated GDE, with (A) carbon paper structure, (B) conduction, (C) flow, and (D) diffusion.

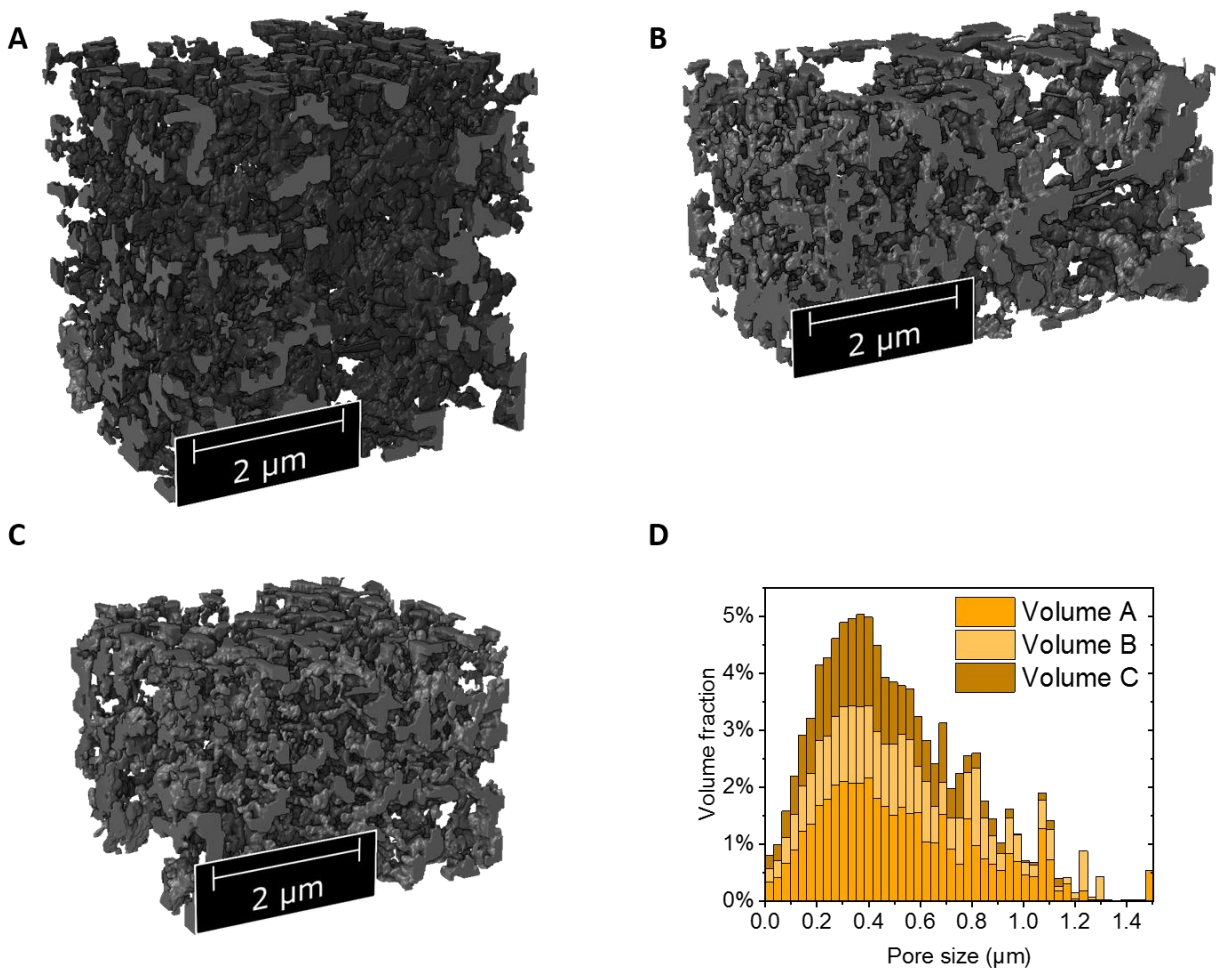


Figure S14 Visualizations of the porous carbon volumes in the MPL (A), (B) and (C), with the pore size distribution (D) indicating their contributions.

Table S6: Equations used for the computation of the transport properties

| | | |
|---|---|---|
| Fick's Law | $\mathbf{j} = -D_{eff}\boldsymbol{\nabla} c$<br>$\Delta c = 0$ | **j:** diffusion flux<br>$D_{eff}$ : effective diffusivity tensor<br>$c$ : the concentration |
| Definition of effective diffusivity | $D_{eff} = D_0\, D_{rel} = D_0\, \frac{\rho}{\tau}$ | $\rho$: porosity<br>$\tau$ : tortuosity<br>$D_0$ : species diffusivity<br>$D_{rel}$: relative diffusivity |
| Stokes equation | $-\mu\Delta\mathbf{v} + \boldsymbol{\nabla} p = 0$<br>$\boldsymbol{\nabla}\cdot\mathbf{v} = 0$ | $\mathbf{v}$: velocity<br>$\mu$: dynamic viscosity<br>$p$ : pressure |
| Darcy's law | $\mathbf{v}_{avg} = -\frac{k}{\mu}\frac{\boldsymbol{\nabla} p}{L}$ | k: permeability<br>L: layer thickness |
| Ohm's law | $\boldsymbol{j} = -\sigma_{eff}\boldsymbol{\nabla}\Phi$<br>$\Delta\Phi = 0$ | $\boldsymbol{j}$ : current density<br>$\sigma_{eff}$ : conductivity tensor<br>$\Phi$ : potential |

| | | |
|---|---|---|
| | $\sigma_{eff} = \sigma_0 \cdot \sigma_{rel} = \sigma_0 \frac{\mathrm{SVF}}{\tau_s}$ | $\sigma_0$ : material conductivity<br>$\sigma_{rel}$ : relative conductivity<br>$\mathrm{SVF}$ : solid volume fraction<br>$\tau_s$ : solid tortuosity |
| | $T = \sum_1^N T_i^{part} w_i$<br>$w_i = \frac{V_i}{\sum V_i}$ | $T$ : transport parameter<br>$N$ : number of subvolumes<br>$T_i^{part}$: subvolume transport parameter<br>$V_i$ : volume of subvolume i |

**Binder distribution in the catalyst layer**

The catalyst layer Nafion is expected to occupy a portion of the pore space. The fact that it is not observed in the FIB/SEM images and tomogram may have several reasons. The binder content in the ink may deviate substantially from the binder content in the final porous structure due to ink stability and drying conditions. The binder may be damaged by the FIB or electron beam, especially if the layers are thin (this is a known issue and can only be partially mitigated by cryogenic processes). The binder may be obscured by the much heavier copper, which is likely to feature much brighter in the images.
Six different models of the catalyst layer with various Nafion weight percentages were generated, and the calculation of the relevant parameters was repeated for each model. The models are based on two different conjectures about the location of the Nafion in the catalyst layer. In principle, both models lead to a thin layer or small deposits congruent with the inability to detect them by FIB/SEM. The first model assumes the entire catalyst surface is coated in a thin layer (one or two voxels, i.e., 8-16 nm) of Nafion. This assumption is congruent with the Nafion distribution encountered in fuel cell catalyst layers.[1] The modeling parameter here is the thickness of the coating layer. The second family of models assumes Nafion to be concentrated in the small pores. The model assigns binder to small pores in the structure until a certain approximate weight percentage is reached. This algorithm is included in GeoDict and dubbed AddBinder there. The modeling parameter is the targeted weight percentage. The base model without Nafion likely overestimates the transport of reactants in the catalyst layer. The true transport parameters are likely lower due to Nafion partially blocking transport pathways. Visualizations of the models can be found in Figure S15. The restrictions to the transport are assessed in Figure S16 for diffusion and Figure S17 for Stokes flow.

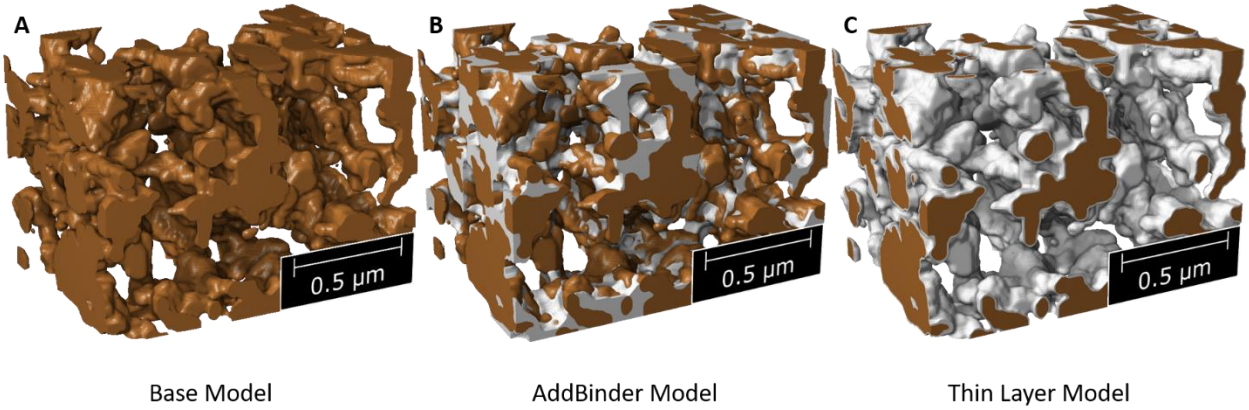


Figure S15 Visualizations of the Nafion distribution models for the catalyst layer. A) shows the base model without nafion, B) show the GeoDict AddBinder model with approximately 5 weight percent Nafion, C) shows the thin layer model with a single voxel coating of Nafion.

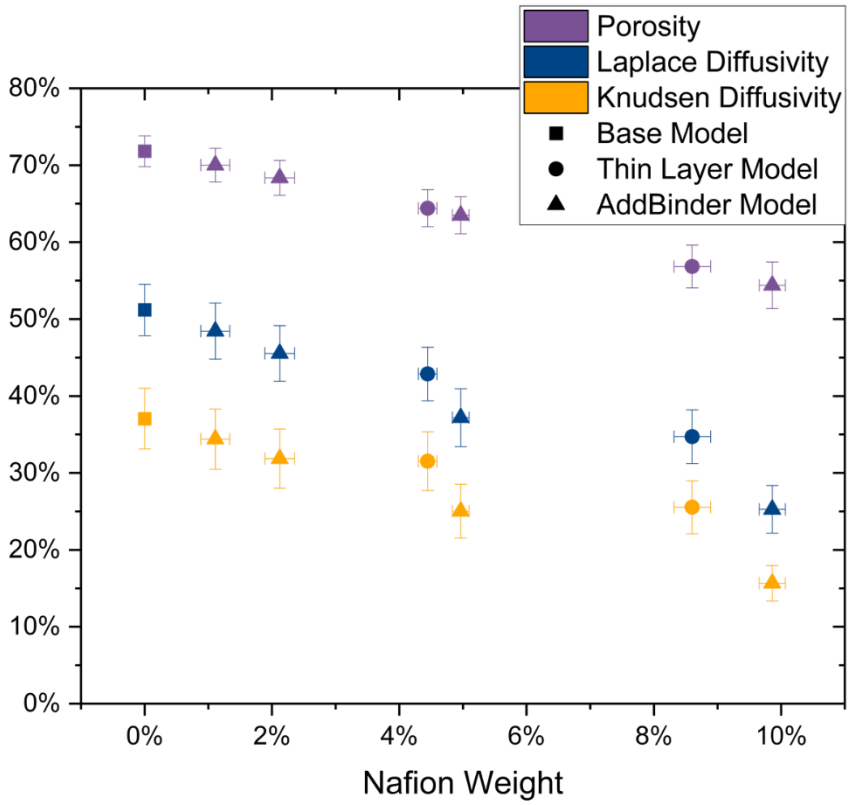


Figure S16 Change in porosity and relative diffusivity with increasing Nafion weight percentage. In all cases only the through plane direction is shown. The diffusivities depend on the choice of model. The thin layer model has a weaker effect on the diffusivity than the AddBinder model. Both models lead to a reduction in the available pore space. The thin layer model leads to constriction, whereas the AddBinder model leads to outright blockage of small pores.

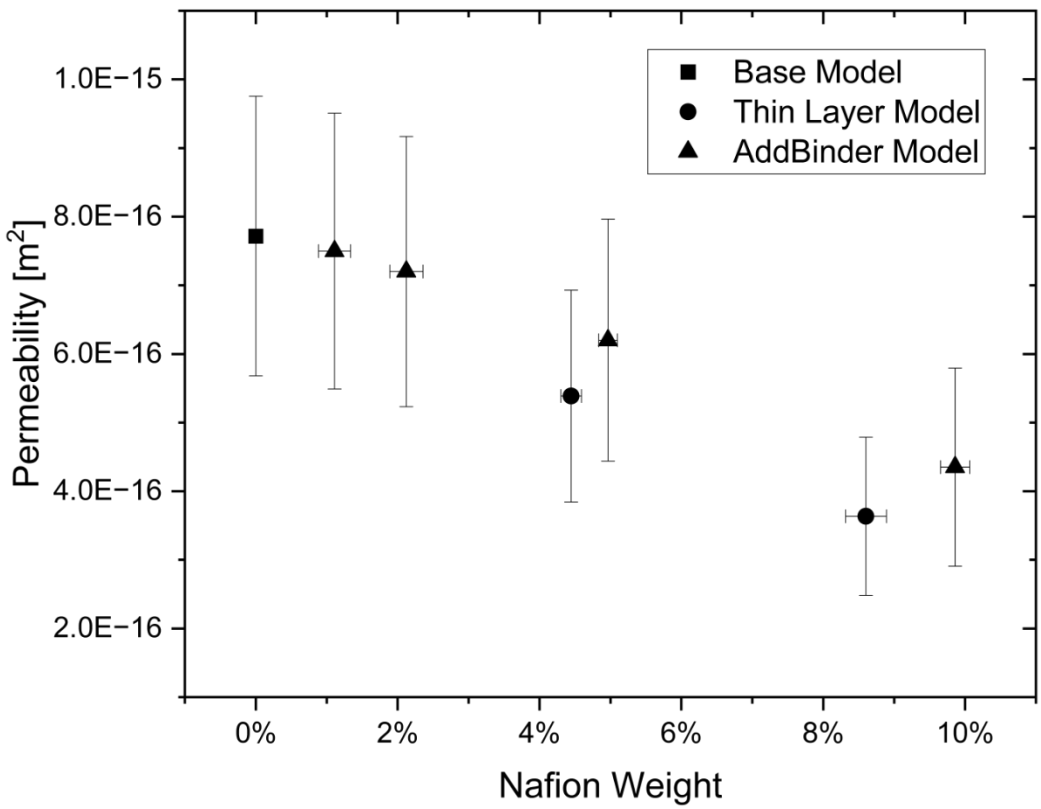


Figure S17 Change in permeability as a function of Nafion weight percentage for the six models. The thin layer model affects the permeability more than the AddBinder model.

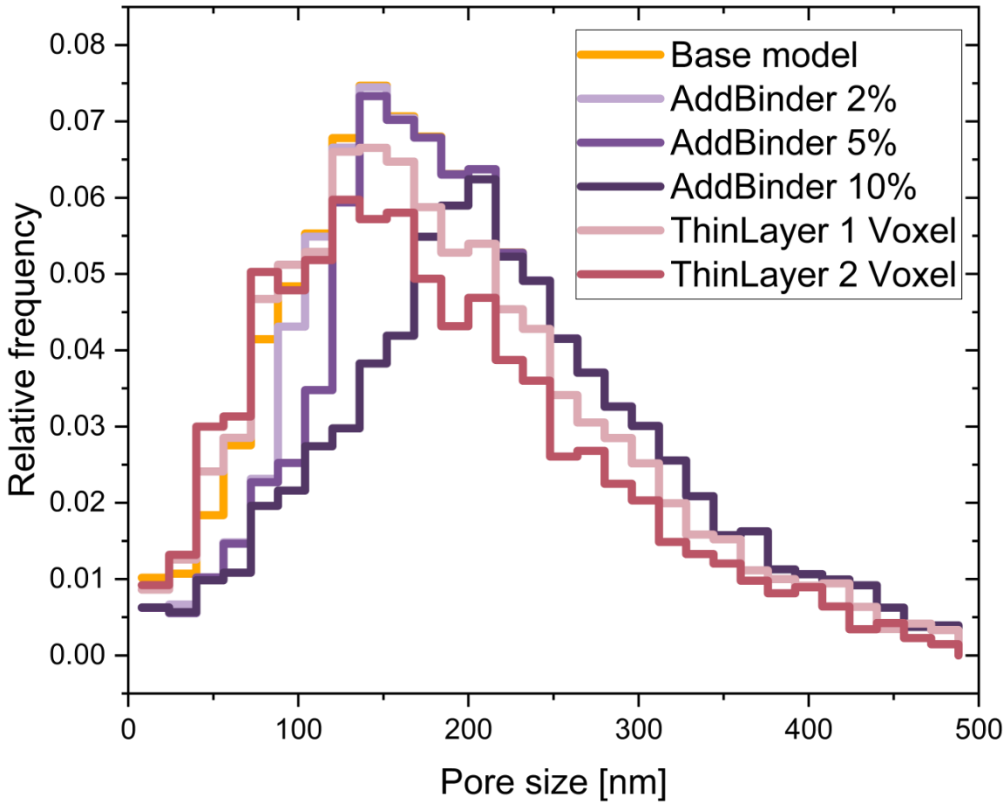


Figure S18 Pore size distribution of the Nafion binder models. The thin layer model reduces the availability of large pores, whereas the AddBinder model leaves large pores intact, but reduces small pore availability.

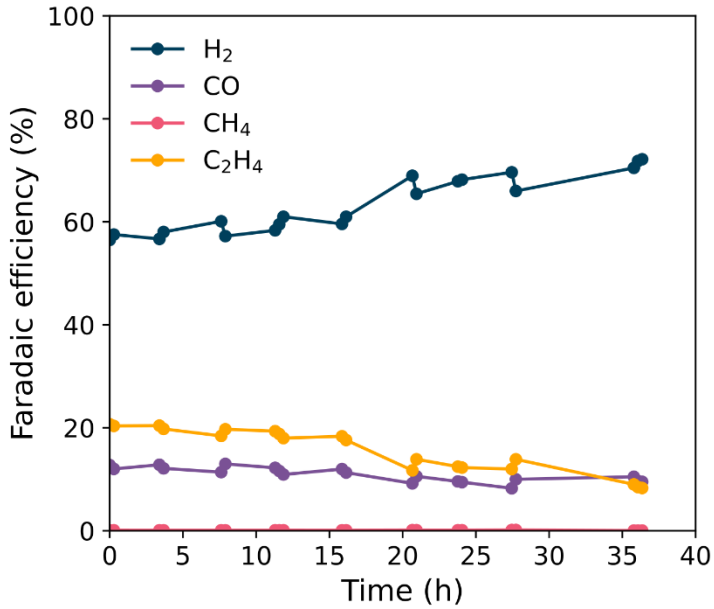


Figure S19 Stability of an 800 $cm^2$ cell with the Freudenberg H23C8 GDL at 125 $mA/cm^2$ (100 A). The $C_2H_4$ FE decreased from 20% to 10% after 37 hours. With the segregated GDL in the 800 $cm^2$ cell, the $C_2H_4$ FE remains above 15% for more than 240 hours.

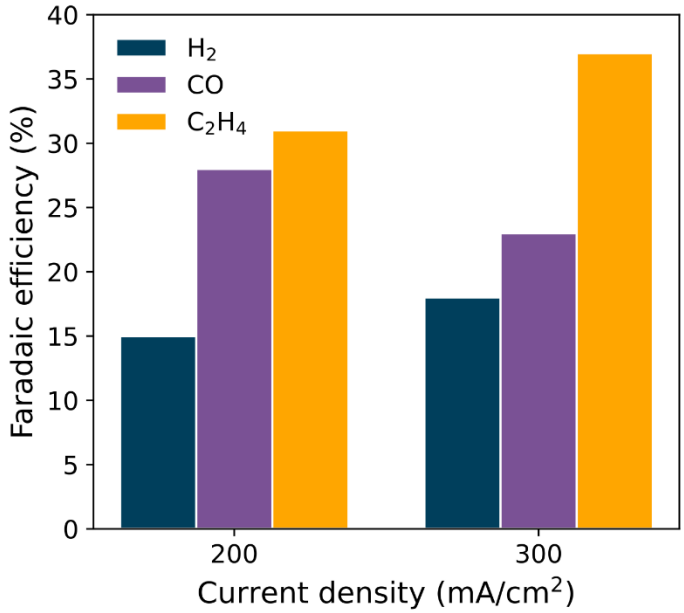


Figure S20 Selectivity of an MEA assembled in the 25 $cm^2$ cell with the segregated GDE.

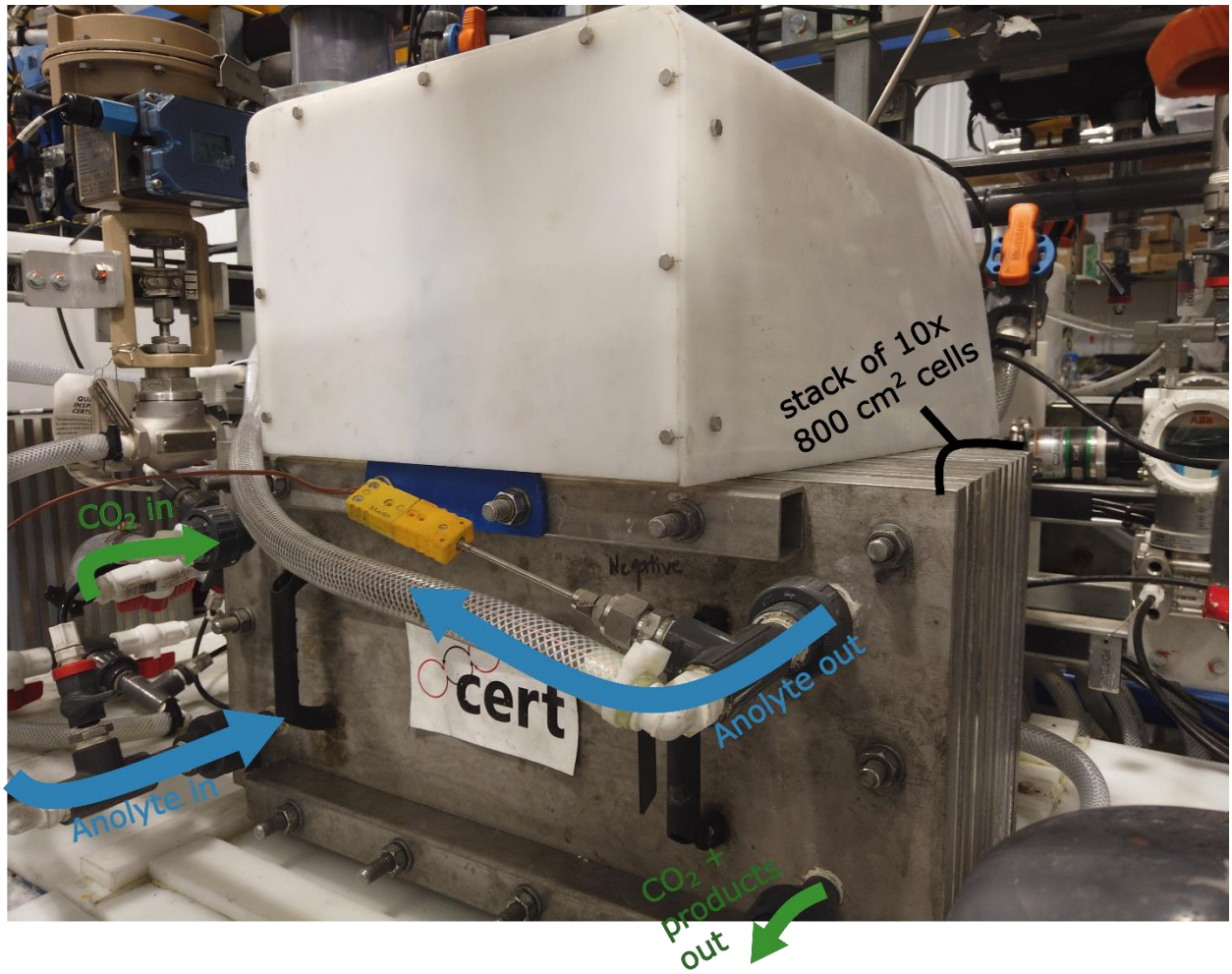


Figure S21 Picture of the 10-cell stack of 800 $cm^2$ cells with annotations of the fluid connections.

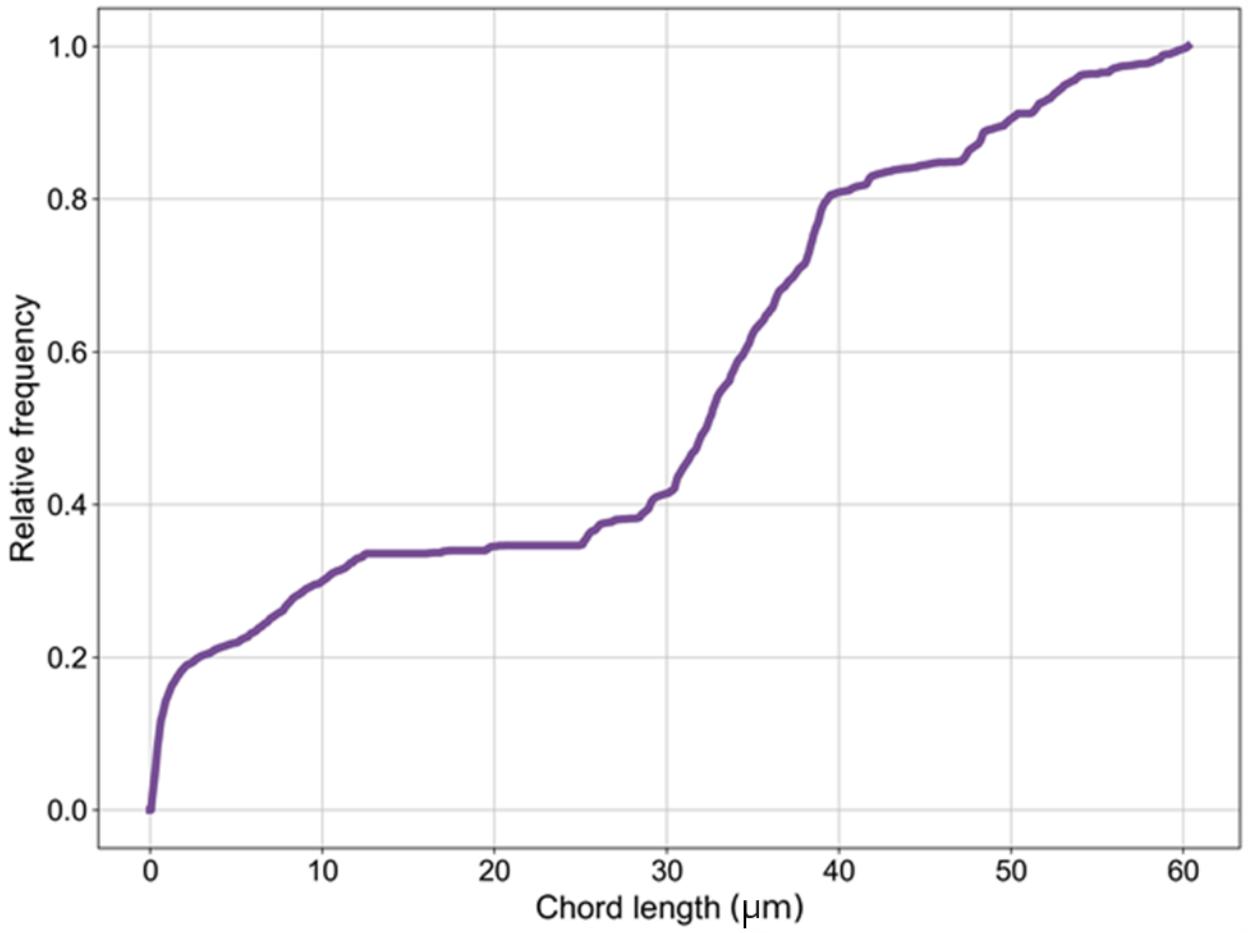


Figure S22 Cumulative chord length distribution estimating the thickness of the segregated MPL. The median chord length is found at 32.2 µm. The distribution was attained by applying a UNET2D AI algorithm to segment a cross section image of the MPL. The resulting binary image was analyzed using the GeoDict chord length algorithm.

1. Lopez-Haro, M., Guétaz, L., Printemps, T., Morin, A., Escribano, S., Jouneau, P.H., Bayle-Guillemaud, P., Chandezon, F., and Gebel, G. (2014). Three-dimensional analysis of Nafion layers in fuel cell electrodes. Nat. Commun. *5*, 1–6. 10.1038/ncomms6229.